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# HydraDG: Governed Context Interventions with Fractal Custody for Agent Experiments

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## Abstract

1 Agent evaluations increasingly depend on context interventions—retrieval, mem-  
2 ory graphs, and prompt contracts—yet published reports often omit null outcomes,  
3 parser failures, and negative cells. We present HydraDG, a governed experimen-  
4 tal framework that binds substantive model invocations, tool calls, and handoffs to  
5 *Fractal Custody Objects* (FCOs) and a *Fractal Custody Graph* (FCG). HydraDG  
6 preregisters hypotheses, freezes case manifests and scorers, records raw prompt  
7 and response hashes, and preserves terminal states including timeouts, abstentions,  
8 and malformed outputs without silent substitution. We report two preregistered  
9 studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval  
10 for failure-prevention endpoints; both terminate as *underpowered* under bounded  
11 case replication, so confirmatory effect claims are not supported while negative  
12 and null evidence remains in custody. We argue that custody-first evaluation in-  
13 frastructure supports reproducible agent science rather than optional logging.

## 14 1 Introduction

15 Large language model agents are evaluated on task success, benchmark scores, and human prefer-  
16 ence [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models,  
17 stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative  
18 and null cells disappear from the public record, later work cannot distinguish *no effect* from *no*  
19 *measurement*.

20 HydraDG addresses this gap with a custody-first experimental contract. Every actor—human oper-  
21 ator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a  
22 handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, in-  
23 put hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity  
24 commitments, not digital signatures unless an authorized signing operation occurred). Probabilistic  
25 model outputs are never treated as authority; they are evidence objects subject to deterministic  
26 scoring and graph append rules.

27 This paper describes the HydraDG framework and reports terminal results from the IC Failure Learn-  
28 ing structured-retrieval falsification lane. We do *not* claim that structured FCG context improves  
29 failure prevention; preregistered EXP-008 and EXP-009 closed as underpowered. We also do not  
30 treat exploratory secondary patterns as promoted findings.

## 31 2 Related Work

32 **Agent evaluation.** Benchmark suites measure capability but rarely require publication of failed  
33 runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks  
34 with mandatory negative capture at the custody layer.

35 **Retrieval and structured context.** Retrieval-augmented generation and graph-augmented memory  
36 systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null  
37 retention, and power-aware verdicts; HydraDG enforces both.

38 **Provenance and reproducibility.** Preregistration, FAIR data principles, and nanopublication-  
39 style lineage support scientific reproducibility [5–7]. Agent stacks often stop at prompt logging.  
40 FCO/FCG extend content-addressed lineage to cross-actor handoffs so that a stale agent cannot  
41 write after ownership moves. Companion custody-framework preprints (deferred to camera-ready  
42 de-anonymization) define FCO/FCG formally; they establish framework provenance, not the Hy-  
43 draDG empirical results reported here.

## 44 3 Framework

### 45 3.1 Custody objects

46 An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call,  
47 scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges  
48 between FCOs with hash-linked roots. HydraDG preserves failures, nulls, and malformed outputs in  
49 custody (a failure-complete *behavior*, not a redefinition of FCO/FCG). Claim ceilings (exploratory  
50 mechanistic falsification, descriptive only, etc.) are declared at preregistration and enforced at pro-  
51 motion time.

### 52 3.2 Experimental freeze

53 Each experiment declares:

- 54 • frozen case manifest and prompt-contract hashes;
- 55 • conditions (e.g., flat prose vs. structured FCG);
- 56 • models with expected runtime digests;
- 57 • primary endpoint and aggregation rule;
- 58 • explicit gates for confirmatory vs. exploratory claims.

59 Execution emits `START_RECEIPT`, per-cell raw outputs with prompt/response SHA-256, parser state,  
60 and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substi-  
61 tution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as  
62 terminal evidence.

### 63 3.3 Governed local execution

64 Where preregistered, local model calls route through a governed bridge that records selection age,  
65 swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate  
66 evidence classes; one is not silently labeled as the other.

## 67 4 Experimental Setup

### 68 4.1 Task family

69 The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints  
70 such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen `CASES.jsonl`  
71 manifest; each case receives multiple replicates per condition.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

| Study   | Primary verdict | Raw cells | Valid parse rate |
|---------|-----------------|-----------|------------------|
| EXP-008 | UNDERPOWERED    | 300       | 0.907            |
| EXP-009 | UNDERPOWERED    | 300       | 0.883            |

## 4.2 Conditions and models

**C0 (FLAT\_PROSE):** failure-handling guidance as unstructured text.

**C1 (STRUCTURED\_FCG):** the same semantic content composed through a frozen FCG retrieval contract.

EXP-008 and EXP-009 used local models qwen3:1.7b and qwen2.5-coder:7b under identical case and scorer contracts. Thinking/reasoning modes were held consistent with preregistration. Primary aggregation: majority of three replicates per case.

## 4.3 AI and agent methodology disclosure

Frontier coding agents assisted with repository tooling and manuscript preparation; they are not listed as authors. All scientific endpoints come from preregistered local model executions with hashed prompts and responses. Routine editing does not constitute a reported experimental intervention.

## 4.4 Power and claim gates

E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary promotion.

## 5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Both experiments executed the full raw matrix ( $n = 300$  cells each) with complete custody append.

**EXP-008.** Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ( $n = 2$  paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

**EXP-009.** Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An exploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is **not promoted** to a causal claim.

**Stage-2 baseline.** Prior IC Failure Learning stage-2 execution (414 scored rows across two models) established FAILURE\_LEARNING\_BEHAVIOR\_IMPROVEMENT\_NOT\_ESTABLISHED, providing historical context for the structured-retrieval falsification lane.

## 6 Discussion

HydraDG shows that agent experiments can retain null and failure cells with the negative-result integrity expected in preregistered experimental sciences: failures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while preserving malformed and timeout cells for audit.

### 6.1 Limitations

Limitations include bounded case replication, local-only model lanes in the reported matrix, and absence of runtime-equivalent successor replication in primary results. A preregistered Qwen 3.8

108 successor lane exists but is non-terminal and omitted from primary results. Whole-project hierarchi-  
109 cal atomization (SeedGraph v1a validation) ran for multiple days but was interrupted during final  
110 Parquet serialization without a BUILD\_RECEIPT; partial node/edge artifacts are not readback-safe,  
111 so terminal whole-corpus atomization is not established in this submission. Biological, protein-  
112 structure, and cellular perturbation applications of the custody framework remain future validation  
113 targets, not demonstrated generalizations of the agent-context results here.

## 114 6.2 Future directions

115 **Agent systems.** Larger successor replications, governed multi-agent handoffs, selective replay, and  
116 resource-aware custody gates under preregistered manifests.

117 **Structured retrieval at scale.** Checkpointed ingest and transactionally durable atomization so long  
118 hierarchy builds can resume from batch boundaries rather than materializing the entire graph at  
119 finalization.

120 **Cross-project federation.** Project-local FCG roots with explicit cross-project references and no  
121 automatic evidence admission between repositories.

122 **Cryptographic custody.** Authorized signatures and MMR commitments where policy permits;  
123 hashes remain identity commitments unless a verified signing operation occurred.

124 All directions above are *planned* unless independently measured; they do not extend the EXP-  
125 008/009 empirical claims.

## 126 7 Conclusion

127 We presented HydraDG, a custody-first framework for agent context-intervention experiments, and  
128 reported two terminal underpowered preregistered studies that block premature promotion of struc-  
129 tured FCG retrieval effects on failure-prevention endpoints. We recommend FCO/FCG custody,  
130 explicit claim ceilings, and terminal-state preservation as baseline infrastructure for NewInML-style  
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